

MTH514A: End-semester Examination

Time: 120 minutes

Answer ANY FIVE of the following questions. If you answer more than five questions, then the first five will be graded. [10 × 5 = 50]

Q 1. Let $C_1, \dots, C_k$ be orthogonal projection matrices such that $C_1 + \dots + C_k = I$ and $X^{n \times p}$ be a data matrix from $N(\underline{\mu}, \Sigma)$. Define $M_i = X' C_i X$ for $i = 1, \dots, k$.

(a) Show that $M_1, \dots, M_k$ are jointly independent. [Hint: Use Fisher-Cochran Theorem]

(b) If $\underline{\mu} = \underline{0}$, then show that $M_i \sim \text{Wishart}_p(\Sigma, r_i)$ for $i = 1, \dots, k$, where $r_i = \text{rank}(C_i)$.

(c) For general $\underline{\mu}$ if $C_1 = n^{-1} \underline{1} \underline{1}'$, then show that $M_i \sim \text{Wishart}_p(\Sigma, r_i)$ for $i = 2, \dots, K$.

Q 2. Let $\underline{X} = (x_1, \dots, x_4)'$ be a sample from Multinomial($n, p_1, \dots, p_4$). Using multivariate CLT and multivariate delta theorem, derive the asymptotic distribution of $\{\log(x_1 x_2) - \log(x_3 x_4)\}$ as $n \rightarrow \infty$.

Q 3. Let $\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ follows $N(\underline{\mu}, \Sigma)$, and $\underline{X}_1, \dots, \underline{X}_n$ be n random samples from $N(\underline{\mu}, \Sigma)$.

(a) Using the Cramér Wold theorem, can you construct a framework for the union-intersection test for testing " $H_0 : x_1 \text{ is independent of } x_2$ ", based on these n samples?

(b) Suppose we set the test statistic for testing independence of two univariate random variables y and z by $r_{y,z}^2$. Then, derive the union-intersection test statistic for testing the hypothesis H_0 stated in part (a).

[Consider taking appropriate assumptions if necessary.]

Q 4. Let $\underline{X} = (X_1, \dots, X_p)'$ follow $N(\underline{\mu}, \Sigma)$, where both $\underline{\mu}$ and Σ are unknown.

(a) Show that the likelihood ratio test (LRT) statistic (Λ) for testing " $H_0 : X_1 \text{ is uncorrelated with } X_2, \dots, X_p$ " based on n random samples is given by

$$\Lambda = \left(1 - r_{X_1 \bullet X_2 \dots X_p}^2\right)^{n/2},$$

where $r_{X_1 \bullet X_2 \dots X_p}$ is the multiple correlation coefficient between X_1 and $X_2, \dots, X_p$.

(b) Hence, or otherwise show that the first sample canonical correlation coefficient between the two sets of random variables $\underline{X} = [X_1, \dots, X_p]'$ and $\underline{Y} = [Y_1, \dots, Y_q]'$ is bigger than or equal to all possible multiple correlation coefficients between $\underline{X}$ and $\underline{Y}$.

Q 5. (a) The residual variance in predicting x_i by linear regression on $\underline{b}'\underline{y}$ is given by

$$\delta_i^2 = \text{var}(x_i) - \text{cov}^2(x_i, \underline{b}'\underline{y}) / \text{var}(\underline{b}'\underline{y}), \quad i = 1, \dots, q.$$

Show that the vector $\underline{b}$ which minimizes $\sum_{i=1}^q \delta_i^2$ is the eigenvector corresponding to the largest eigenvalue of $\Sigma_{y,y}^{-1} \Sigma_{y,x} \Sigma_{x,y}$.

(b) Show further that the best k uncorrelated linear functions of $y_1, \dots, y_p$ for predicting $x_1, \dots, x_q$ in the sense of minimizing the sum of squared residual variances correspond to the first k eigenvectors of $\Sigma_{y,y}^{-1} \Sigma_{y,x} \Sigma_{x,y}$.

Q 6. (a) Let R be a 3×3 correlation matrix with $r_{1,2} = r_{1,3} = 1/3$ and $r_{2,3} = 1/10$. Show that R is positive definite.

(b) Suppose we write $R = \underline{\lambda} \underline{\lambda}' + \Psi$, where $\underline{\lambda} \in \mathbb{R}^3$ and Ψ is a diagonal matrix. Find the exact factor solution of $\underline{\lambda}$ and Ψ . Further, show that this solution is unacceptable.

Q 7. Let $\mathbf{x}$ follow a p -dimensional elliptical distribution with density function

$$f_{\mathbf{x}}(\mathbf{x}) = \det(\Sigma)^{-1/2} g \left[(\mathbf{x} - \mu \mathbf{1})^T \Sigma^{-1} (\mathbf{x} - \mu \mathbf{1}) \right],$$

where $\Sigma = \sigma^2 [(1 - \rho)I + \rho \mathbf{1}\mathbf{1}^T]$. Find an appropriate orthogonal transformation $\mathbf{y} = H\mathbf{x}$, such that $\mathbf{y}$ has the density

$$f_{\mathbf{y}}(\mathbf{y}) \propto g \left[c_1 (y_1 - \sqrt{p}\mu)^2 + c_2 \sum_{j=2}^p y_j^2 \right],$$

where c_1, c_2 are constants. Also, find the values of c_1, c_2 and the proportionality constant in $f_{\mathbf{y}}(\mathbf{y})$.

Q 8. Let Σ be a $p \times p$ positive definite covariance matrix, and consider the problem of approximating Σ by a $p \times p$ matrix M of rank k obtained by minimizing

$$\|\Sigma - M\| = \left[\sum_{i=1}^p \sum_{j=1}^p (\sigma_{i,j} - m_{i,j})^2 \right]^{1/2}.$$

Show that

$$\|\Sigma - M\|^2 = \text{trace} \{ (Q - \Lambda) (Q - \Lambda)' \},$$

where $\Sigma = \Gamma \Lambda \Gamma'$ (the diagonal elements of Λ are ordered) and $Q = \Gamma' M \Gamma$. Hence, show that the matrix M of rank k , which minimizes $\|\Sigma - M\|$ is the k -PC projection matrix.

All the best!